Pencil upper gate: terminal checkpoint 18 observations

Run: pencil-p186379-20261030-c1. Caller-supplied source commit: 543a54b4b1c3be7052872af3ffe393a44c69a3c3.
Observations only. No statistical, speed, mathematical or timing qualification is inferred.
One public synthetic mathematical target at p=186379. Controller failed during checkpoint19; last committed checkpoint18 is incomplete.
Workloads: 1; expected pairs per price cell: 5; slots: 18; observed: 7; missing: 11.

Recorded outcomes

pencil:timeout: 2
rho:verified: 3
rho_AA:verified: 2
Missing rows remain missing; capped, timeout, error and other outcomes retain their original labels. Counts from different epochs are not pooled.

Pair coverage by price cell

Workloads: 1; expected pairs per price cell: 5; slots: 18; observed: 7; missing: 11.

Arms: pencil, rho, rho_AA. Planned slots are not observations.

Fewer than two comparable ratio cells are present. Diagram of planned scope and observed progress; no performance plot.
Observations only. No algorithm ranking, mathematical verdict, timing qualification or successful candidate/rho ratio.
Last committed generation18 archive and selected session files passed versioned hash and size checks. Embedded session remains as recorded; its running metadata is not rewritten.
Seven of18 planned slots are retained: five producer-verified rho/control rows and two pencil watchdog timeouts, one warmup and one measured. Producer verification is not independent replay.
Three warmups and four measured executions are observed;11of15 measured slots are missing. No successful candidate exists. Missing work is not zero cost.
Controller3 failed at07:54:08.858UTC with CRIU dump failed; retained log, no generation committed. Generation19 is not an additional observation.
Scientific protocolv4 and unchanged instrument govern rows. Version5 changed only controller supervision. No fourth controller or replacement run occurred.
All clocks remain unqualified. Total active worker time and spending remain unknown. Prior failed campaign costs remain separately bound.
Canonical package/check retain incompleteness findings without suppression. Independent terminal review is pending because delegated agents reported a usage limit.
The six-hour authorization expired at11:17:54UTC; this saved-record analysis authorizes no new computation. Full1260 pencil evidence remains incomplete.

Separate resource observations

No conversion to algorithm cost units.


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{"accounting_notes":["Controller terminal; generation19 failed to commit, generation18 retained.","Two pencil ti
meouts are retained, including one measured row; no candidate solve.","Active worker time, exact termination ti
me and total spend remain unknown.  EC2 lookup returned no instance, not an exact termination receipt.","No ind
ependent replay or new experiment was launched; expired budget is not renewed.","Prior b1 failed attempt costs
remain linked separately, without refund or pooling."],"allocated_dollars":null,"allocation_and_algorithm_costs_d
istinct":true,"charged_categories":["candidate","rho","rho_AA","warmups","computation","fresh_epoch_if_any","inde
pendent_replay_if_any","recovery","failed_attempts"],"controller_amendment":{"path":"/home/ec2-user/crypto-pencil
-signed-graph-20261029/research/pencil_large_prime_20261029/protocol_draft/replacement_upper_gate_20261030/protoc
ol/controller-continuation-v5.json","sha256":"d00be8a9e4b12fd4e6bb60d3d6c423b9213850de1859e22e248b52cddf70c956"
},"controller_build":"isolab-recovery-controller:2fe8ec79-1f54-4abe-8ec4-151b50b1178b","controller_continuation_
admission":{"path":"/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/prot
ocol_draft/replacement_upper_gate_20261030/controller-takeover/continuation-admission.json","sha256":"187f2154428
0fdef4835b52e1ea860a1ed8472e0f3b50b2a6b6fdfd5da215ca2"},"controller_end_utc":"2026-10-30T07:54:08.858Z","control
ler_history":{"path":"/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/run
```

s/p186379-gate-20261030-c1/terminal-g18-observations/terminal-build.json","sha256":"52baba6d7bca8730f3d4c56d0559e2dd7b645068378ce38c6ce4de226760d31a"},"controller_invocations_observed":3,"controller_phase":"COMPLETED","controller_status":"FAILED","cost_conversion":null,"cumulative_active_worker_seconds_limit":21600,"failed_costs_retained":true,"maximum_active_scientific_payloads":1,"missing_costs_not_zero":true,"observation_utc":"2026-10-30T14:46:00Z","original_authorization_start_unix":1793337474,"outer_D_unix":1793359074,"prior_b1_resources":{"path":"/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-b1/terminal-partial/native-analysis/resource-observations-worker-terminal.json","sha256":"4880cc449b6d4b0065cd71d1be119cb748bb5f658d8574b87051454738b6f928"},"request_G_unix":1793347824,"resource_authorization":{"path":"/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/protocol_draft/replacement_upper_gate_20261030/resource_authorization.json","sha256":"5ea8e080af4b28572f3ff1d074e9e261ef8ad3c24a53a6e7d06d1b37b0316d25"},"schema":"pencil-terminal-resource-observations/v1","scientific_submission_count":1,"scientific_total_cost":null,"scope":"Terminal controller and last committed checkpoint; scientific accounting incomplete. Allocation is not active execution time.","terminal_failure":"CRIU dump failed; retained log, no generation committed","terminal_state":{"path":"/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/terminal-g18-observations/terminal-observed-state.json","sha256":"0c082596e8aa9b5f4b7179c2be6e38eab51123d4de534ab02cf81ab92bfc3307"},"worker_active_seconds":null,"worker_attempt_observed":0,"worker_elapsed_seconds":null,"worker_instance_id":"i-04d58c2b309fb6549","worker_instance_terminated_verified":false,"worker_pod_uid":"3a7ab215-49d1-4982-8c78-5a0c71646842","worker_restores_observed":0,"worker_termination_verified":false}

All36 exact price cells

S2 is twice S/M; N10 is ten times the native event price. Incomplete/unavailable ratios are null, not zero. Partial admitted sums are retained in JSON as diagnostic only; no partial sum is promoted to a complete ratio.

Cell	S2/I/N10	Admitted/expected	Method/rho exact	With calibration/rho exact
1	0/1/0	0/5	null	null
2	0/1/1	0/5	null	null
3	0/1/10	0/5	null	null
4	0/10/0	0/5	null	null
5	0/10/1	0/5	null	null
6	0/10/10	0/5	null	null
7	0/100/0	0/5	null	null
8	0/100/1	0/5	null	null
9	0/100/10	0/5	null	null
10	0/1000/0	0/5	null	null
11	0/1000/1	0/5	null	null
12	0/1000/10	0/5	null	null
13	1/1/0	0/5	null	null
14	1/1/1	0/5	null	null
15	1/1/10	0/5	null	null
16	1/10/0	0/5	null	null
17	1/10/1	0/5	null	null
18	1/10/10	0/5	null	null
19	1/100/0	0/5	null	null
20	1/100/1	0/5	null	null
21	1/100/10	0/5	null	null
22	1/1000/0	0/5	null	null
23	1/1000/1	0/5	null	null
24	1/1000/10	0/5	null	null
25	2/1/0	0/5	null	null
26	2/1/1	0/5	null	null
27	2/1/10	0/5	null	null
28	2/10/0	0/5	null	null
29	2/10/1	0/5	null	null
30	2/10/10	0/5	null	null
31	2/100/0	0/5	null	null
32	2/100/1	0/5	null	null
33	2/100/10	0/5	null	null
34	2/1000/0	0/5	null	null
35	2/1000/1	0/5	null	null
36	2/1000/10	0/5	null	null

Evidence bindings

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/terminal-g18-observations/native-analysis/native-analysis.json
32708df2c828386607f786418e464b88858f38c3e8a234a04fa808a946ab05d7
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/terminal-g18-observations/observation.json
89ed5c95d28c8981bace4a76f3839cad2c821d2a5de3755a8ce7b996af6f84f3
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/terminal-g18-observations/terminal-observed-state.json
0c082596e8aa9b5f4b7179c2be6e38eab51123d4de534ab02cf81ab92bfc3307
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/terminal-g18-observations/native-analysis/execution-history.json
eca752e3f732dc02b9d5120febf01731e257b4e1d31d49e9ff87e623e7c229f3

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/terminal-g18-observations/native-analysis/export.json
40836bd4cd394adcca6074922352af08b131c9922bd33ab7e251d25497319fdd
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/terminal-g18-observations/native-analysis/resource-observations.json
5a6f190e1677b3cc8e2bf8e885bf1c7ecfb54195a01849f79e62675ff3193dd2